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LARSON—MATH 756—CLASSROOM WORKSHEET 07
Hall and König's Theorems

Review

1. What is Hall's Condition?
2. What is the “easy” direction of Hall's Theorem?

New

Hall's Theorem. Let G be a (finite) bipartite graph with partite sets A and B . If, for every $X \subseteq A$, $|N(X)| \geq |X|$ then there is a matching in G which saturates A .

Proof. Let M be a matching in G , and let $M_A \subseteq A$ be the vertices in A saturated by M , and $M_B \subseteq B$ be the vertices in B saturated by M . If M does not saturate A then there is an M -unsaturated vertex $a \in A$. Let C be the vertices reachable from a by an M -alternating path (with $C_A = C \cap A$ and $C_B = C \cap B$).

Since each vertex v in $C_B \cap M_B$ is matched to a unique vertex v' in $C_A - a$ with $\{v, v'\} \in M$ (as each vertex in $C_A - a$ is incident to exactly one edge in M), we have $|C_A - a| = |C_B \cap M_B|$. Since $|N(C_A)| \geq |C_A| = |C_A - a| + 1 = |C_B \cap M_B| + 1$ there must be at least one vertex b in $C_B \setminus M_B$ and an M -alternating path from a to b whose last edge is a non- M edge.

Let P be an M -alternating path from a to b , let M_P be the edges in P that are in M (so $M_P = P \cap M$) and let $\overline{M_P}$ be the edges in P that are not in M (so $\overline{M_P} = P \setminus M_P$). So $|\overline{M_P}| = |M_P| + 1$. Now let $M' = (M \setminus M_P) \cup \overline{M_P}$. So $|M'| = |M| + 1$. That is, M' is a larger matching. We see that, for any matching that does not saturate A , we can construct a larger matching and that, by iterating this algorithm, we must terminate with a matching that saturates A .

1. Why is the “hard” direction of Hall's Theorem true? (and what is the main proof idea)?

2. What is König's Theorem?

3. What is a *Min-Max Theorem*? What is an advantage of a min-max theorem?

4. What is the “easy” direction of König's Theorem?

5. Why is the “hard” direction true? (and what is the main proof idea)?

6. What is a Vertex Packing Integer Program?